

Crack Spread

~ OK, stop the giggling. It's more serious than the title suggests. Sit up straight.

David L. Passmore, <https://davidpassmore.net>, 26 August 2026

In petroleum refining, a *crack spread* is a term that can estimate the profitability of crude oil refining for energy companies and their investors, signal supply/demand conditions for oil producers and public and private energy market analysts, and even explain to us pitiable consumer chumps why, for example, gasoline and diesel prices can rise even when crude oil prices are falling.

WHAT IS A CRACK SPREAD?

Crude oil is not used directly. Long-chain hydrocarbon molecules in crude oil are *cracked* using heat, pressure, and catalysts into short-chain hydrocarbons such as gasoline, heating oil, propane, diesel, jet fuel, or other products used in homes, industries, and transportation. As depicted in Figure 1,¹ petroleum products are refined by fractional distillation of crude oil, a process that separates lighter products (e.g., petroleum gas, gasoline) from heavier products (e.g., lubricating oil, bitumen) according to their boiling points.

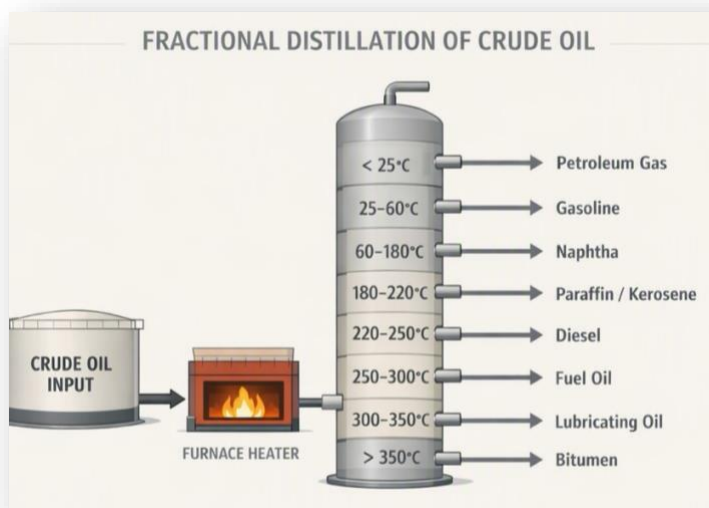


Figure 1. *Cracking Crude Oil*

¹ I created the image in Figure 1 from a prompt I submitted in Google Gemini and Figure 3 as an infographic in Google Notebook based on seven text and one video source. I created a brief video in Google Notebook referenced in this essay using the same eight sources to explain the term “crack spread” (Passmore, 2026). I did not use LLMs or other AI tools to prepare any other part of this essay.

The *crack spread* is the difference between the market value of refined petroleum products and the cost of the crude oil used to make them. Wider crack spreads show that refined products are expensive relative to crude, a condition that can benefit refiners. Narrower crack spreads indicate weaker product prices or more expensive crude, reducing refining profitability. The crack spread, usually expressed in dollars per barrel, is a rough proxy for the gross refining margin between the cost of the crude oil and the value from selling refined products.

But accounting for a crack spread is more difficult than calculating it. In Table 1 are shown possible reasons behind a crack spread widening or narrowing. Inventories of refined products, customer demand, refinery conditions, and changes in crude or refined prices can cause the crack spread to widen or narrow. Calculating the spreads requires analysis of available production and market data. It's just spreadsheet stuff. Even more, though, tactics for responding to a crack spread require a more holistic view of politics, regulations, status and markets for refining inputs, and customer behavior, along with strategic wisdom about energy futures.

Table 1. *Accounting for a Crack Spread*

<i>The spread can widen when....</i>	<i>The spread can narrow when....</i>
Fuel inventories tighten	Product inventories build
Gasoline or diesel demand strengthens	Fuel consumption softens
Refinery maintenance or outages cut fuel output	Refinery production increases
Crude falls faster than product prices	Crude rises faster than product prices

Source: (Carnegie, 2026)

The crack spread is not necessarily profit for the refiner, however. The spread by itself fails to account for costs incurred between oil production and refining—such costs as: energy, labor, transportation, and storage; environmental compliance; the effects of incoming crude on refinery yields; refinery maintenance; and losses or gains resulting from timing refining to the vagaries of the macroeconomy.

AN EXAMPLE: THE 3-2-1 CRACK SPREAD

The *3-2-1 crack spread* is a widely used benchmark, especially in U.S. Gulf Coast refineries. The assumption for this measure is that 3 barrels of crude oil yield 2 barrels of gasoline and 1 barrel of distillate (e.g., diesel/heating oil). This mix approximates a common refinery product rather than the exact mix of any individual refinery. The 3-2-1 common metric defines a hypothetical that allows crack spread comparisons over time and other factors. Of course, calculation of a crack spread is possible for any other refinery product and even for individual refineries and refinery runs within the same facility.

One barrel holds 42 gallons, so prices of crude, gasoline, and distillate are first converted to barrel equivalents:

$$3-2-1 \text{ spread} = [(2 \times \text{gas price} \times 42) + (1 \times \text{distillate price} \times 42) - (3 \times \text{crude price})] \div 3$$

Suppose that crude = \$70/bbl, gas = \$2.50/gal, and distillate = \$2.80/gal.² Then,

- 2 bbl gas: $2 \times \$2.50 \times 42 = \210
- 1 bbl distillate: $1 \times \$2.80 \times 42 = \117.60
- 3 bbl crude: $3 \times \$70 = \210

and, the crack spread: $[(\$210 + \$117.60 - \$210) \div 3] = \$39.20/\text{bbl}$ before accounting for any refinery cost.

Refined products have their own supply-and-demand dynamics, so they do not have to move in step with oil prices. Seasonal demand also affects crack spreads. In North America, domestic gasoline consumption declines in fall and winter months, while distillate consumption rises due to fall harvests and winter heating. May-September is driving season. Refinery maintenance occurs March-May.

Shown in Figure 2 is a plot of the weekly U.S. 3-2-1 crack spread for the year previous to August 14, 2026. The spread has increased steadily since February 2026, peaking in July and August in the \$55 to \$65 range. Of course, the widening gap in the spread tracks with patterns and incidents in President Trump's war of choice against Iran.

² Example from Carnegie (2026).

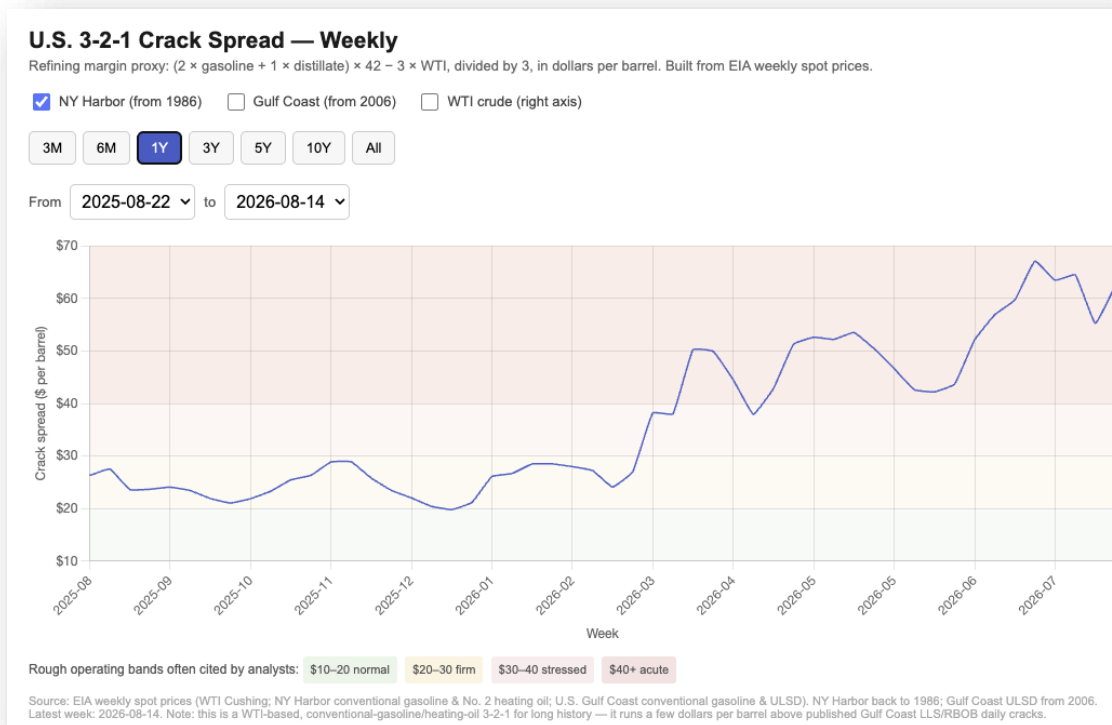


Figure 2. *Weekly U.S. 3-2-1 Crack Spread, August 2025-2026 (Pachman, 2026)*

Shown in Figure 3 is a summary of the “what, why, and how” of crack spreads. A brief video explainer of how the 3-2-1 crack spread works is available (Passmore, 2026).

ARE OIL OR REFINERIES MORE IMPORTANT IN REFINED PRODUCT PRICES?

Consider a situation in which the supply of oil is greater than demand, but gasoline and diesel prices rise. The crack spread is likely to be high because low crude prices would create a gap relative to high pump prices. You might expect refineries to respond by fulfilling consumer demand for fuel to capture premium prices. Not necessarily.

Here is a real scenario that plays out this apparent anomaly. As of early 2024, the U.S. has a refining capacity of approximately 18.4 million bbl/day. However, the industry faced significant challenges, including the loss of 1.1 million bbl/day of capacity due to the COVID-19 pandemic and other factors. Refineries could not reach production capacity no matter how favorable crude/refined product prices became.

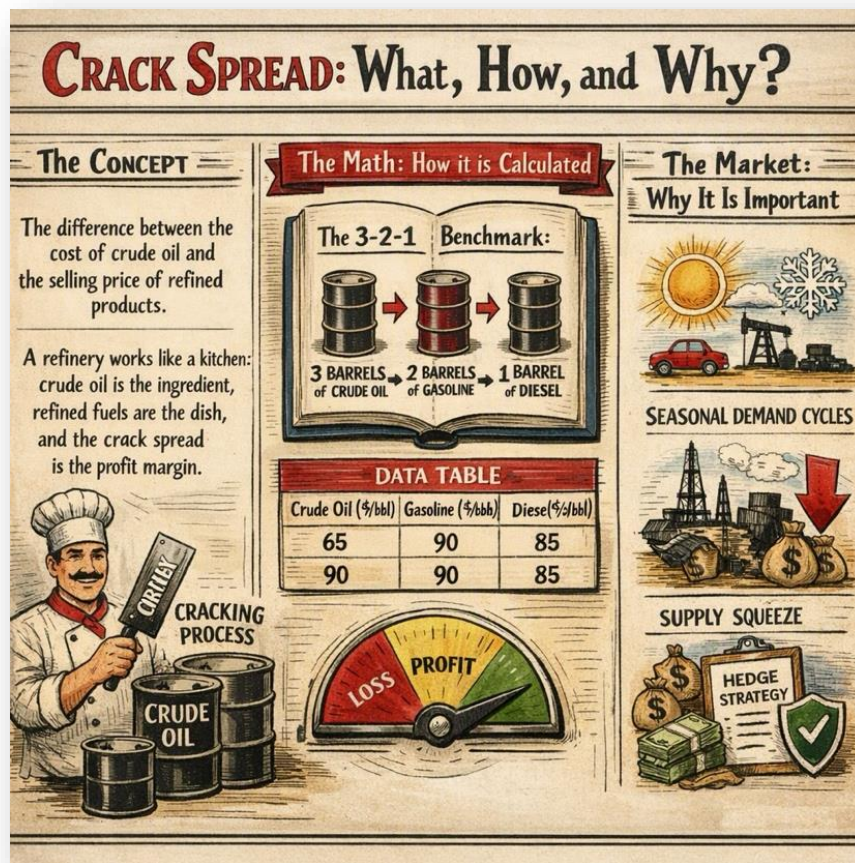


Figure 3. Concept, Calculation, and Importance of Crack Spread

The recent kinetic destruction resulting from Putin’s invasion of Ukraine and Ukraine’s retaliation, as well as from President Trump’s war of choice against Iran, has limited worldwide refining capacity. These wars have driven approximately 9% of global refining capacity offline. Also, Ukraine’s drone bombing campaign is on track to decrease Russia’s refining capacity by 28% over the year. In a *New York Times* op-ed article titled “The ‘Crack Spread’ is Going to Make Your Life Unpleasant,” York (2026) writes that “The world isn’t running out of oil. It is running out of places to turn that oil into fuel. Which means the price of refined products such as gasoline will remain high.”

Other factors can affect capacity. Some refineries in the U.S. have been shuttered or repurposed for renewable fuel production, which reduces the available capacity for traditional petroleum refining. Environmental regulations and permitting processes can hinder the construction of new refineries or the expansion of existing ones.

Further complicating matters is the *rocket-and-feather effect*, also known among frowning professors of economics as *asymmetric price transmission*. It boils down to this asymmetry: prices/wages can rise quickly in response to market conditions, but they are reluctant to fall despite market signals that would support price/wage adjustments. Rocket up. Feather down.

Workers might gladly accept wage gains but might not volunteer for wage cuts when their employers face financial stress. As another example, gasoline prices often shoot up overnight when global oil prices spike but take weeks to drift down when they fall. Adjustments do not mirror the speed of cost changes.

In retail markets, the rocket-and-feather lag often benefits producers at consumers' expense. Sometimes cost increases—or their threat—are passed on in full or even are amplified, but consumers see no relief when producers' costs decrease. Or perhaps prices that rise during a disaster persist after the calm returns. Prices/wages are often described as “sticky.”

CONCLUSION

Monitoring the daily levels and variations in the spot price of a barrel of West Texas Intermediate or Brent Crude will not reveal the future prices consumers will face for gasoline, diesel, home heating oil, jet fuel, or propane. Examining the crude spread, with a little simple reasoning, will provide a more complete picture of the anticipated energy challenges and opportunities. Some links to data series about daily crude spread:

- Y Charts - https://ycharts.com/indicators/brent_wti_spread
- Data4ThePeople - <https://www.data4thepeople.com/p/crack-spread-chart>
- Oil Price API - <https://www.oilpriceapi.com/crack-spread>
- Energy Datasets - <https://www.energydatasets.org/charts/brent-wti-spread>

Focusing on crude oil prices alone is a misleading diversion from a full understanding of petroleum product prices. York (2026) explained how President Trump exploits this folly to roil equity markets:

When President Trump announces yet another potential peace agreement with Iran, crude oil prices fall, and Americans make the mistake of thinking that gas and diesel prices will follow. But crude oil isn't the right barometer of your pain at the pump. By the end of July, crude oil prices had fallen about 25 percent from their April

peak, while gasoline prices dropped only about 9 percent after peaking at around \$4.50 a gallon in May. With prospects for peace fading again, both gas and diesel resumed their upward path, with gas at \$4.07, while diesel reached \$5.47 nationally.

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